

Diagnosing and Correcting NaN Cascades in Learned Dirichlet–Neumann Operator Rollouts

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Abstract

Accurate one-step approximation of a water-wave Dirichlet–Neumann operator (DNO) did not prevent rare catastrophic failures in long symplectic rollouts. Frame-aligned trajectory analysis shows why: the learned operator is accurate on exact solution states but has an incorrect derivative transverse to the solution manifold. It consequently violates the Hadamard shape-derivative identity, so the two equations advanced by the Gauss–Legendre integrator are not the canonical derivatives of one learned Hamiltonian. Small state errors then excite spurious same-sign spectral sidebands, quadratic aliasing amplifies the already established cascade, and the implicit stage iteration eventually loses contractivity. This report summarizes the experimental path that ruled out data corruption, time-step refinement, spectral caps, hard cutoffs, and targeted fine-tuning as primary cures. It motivates and evaluates a clean retraining of $G_0 + G_1 + \mathcal{R}_\theta$, where the analytic first-order Craig–Sulem term is exact, $\mathcal{R}_\theta = O(\eta^2)$ is self-adjoint, and a randomized Hadamard loss directly constrains the missing shape derivative. That structural retraining removed nonfinite predictions on the fixed 320-rollout suite. Subsequent mode-balanced training and a one-epoch stronger tangent continuation produced C21, which remains the best aggregate checkpoint to date: all 320 predictions are finite and the median, 95th percentile, and mean final-elevation errors over 305 truth-valid cases are 0.006866, 0.15369, and 0.03205. The residual tail is no longer a NaN cascade; it is predominantly secular translation error. We derive the phase-sensitive quantity $\gamma^2 = \int B_h \delta c_h^2 dx / \int \eta^2 dx$, show why the existing relative-speed normalization underweights narrow shallow waves, and evaluate a controlled C20-to- γ^2 continuation. This C22 test strongly improves the most phase-sensitive rigid case but fails the pre-registered multi-crest and aggregate-median gates, so it is accepted as partial mechanism evidence and rejected as a replacement checkpoint. Particular care is taken to separate direct measurements, mathematical consequences, and hypotheses still requiring an intervention test.

1 Reader’s guide and executive conclusions

The project accumulated many experiments because the terminal symptom—a NaN inside an implicit time step—admits several superficially plausible explanations. A reader should not be expected to remember the experiment labels. The labels C1–C22 are therefore treated here only as reproducibility handles. The scientific argument is organized around five questions:

1. Is the reference data corrupt or missing the failing regime?
2. Does the learned DNO first make an error on an exact physical state, or does it react incorrectly to the small off-manifold state error created by its own rollout?
3. Is the cascade initiated by the DNO, by pseudospectral aliasing, or by nonconvergence of the implicit Gauss–Legendre stages?
4. Which exact property of a physical DNO is absent from the learned map?
5. What is the smallest architectural and loss-level correction that restores that property without suppressing legitimate high-frequency dynamics?

The present answers, with their evidentiary status, are:

Directly measured:

The failing models can have relative DNO value error of only 2×10^{-4} on the exact state while their response to the difference between the exact and surrogate states grows to a secant gain of 7–8. The induced response first grows in $k = 32, \dots, 64$, then contaminates $k = 64, \dots, 128$, and only afterward makes the quadratic right-hand side strongly aliased and the fixed-point stage solve noncontractive.

Directly measured structural defect:

At the same exact states, the learned operator violates the DNO Hadamard shape-derivative identity by relative $O(1)$, whereas the order-six reference implementation satisfies the identical discrete test to approximately 10^{-5} or better. This discrepancy is far larger than the DNO value error and is already present before visible blow-up.

Mathematical consequence:

If a learned DNO is self-adjoint in ξ but fails the shape identity, then the implemented formulas for η_t and ξ_t are not the two variational derivatives of one learned Hamiltonian. A symplectic integrator cannot restore a Hamiltonian structure absent from the vector field it is given.

Strong mechanistic evidence:

The first Craig–Sulem term has a same-sign Fourier null form that cancels the specific high-mode sidebands observed in the failures. The learned operator does not preserve this cancellation. Exact $G_0 + G_1$ gives 0/32 Tanaka NaNs but is not accurate enough at final time, strongly motivating it as a stable backbone rather than a complete surrogate.

Completed structural test:

The full-capacity $G_0 + G_1 + O(\eta^2)$ model with direct Hadamard-consistency training removed the nonfinite mechanism: the clean C16 epoch-39 checkpoint produced 0/320 nonfinite trajectories on the fixed ten-family suite. Its remaining Tanaka errors were finite phase errors, not renewed spectral cascades.

Current best aggregate checkpoint:

C21, a one-epoch low-learning-rate continuation of the mode-balanced C20 checkpoint with a tenfold larger localized-tangent coefficient, also produces 0/320 nonfinite trajectories and zero counted final divergences. Relative to C20, its full-suite median, 95th percentile, and mean errors improve by 4.66%, 4.43%, and 7.47%, respectively. It repairs the former sole error above one, but worsens a two-crest differential-phase case. Thus a larger mean tangent coefficient is useful but not a complete phase treatment.

Completed phase-normalization ablation:

The current tangent loss measures fractional local speed error, whereas a finite-horizon elevation metric weights the same error by the translation sensitivity $\kappa = \|\eta_x\|_2/\|\eta\|_2$. Replacing the old per-state normalization by the corresponding instantaneous relative-error growth γ^2 is therefore mathematically justified. C22 tests this change in isolation. It reduces case 27 to raw/aligned error 0.18047/0.02688 and preserves 0/64 nonfinite Tanaka predictions, but case 24 remains at 0.63952/0.23662 and the combined raw and aligned medians regress relative to both C20 and C21.

This hierarchy matters. The original operational target of zero nonfinite rollouts has been met on the registered suite, but zero nonfinite values is not the same as uniformly accurate dynamics. C21’s remaining errors distinguish the solved transverse-instability problem from a neutral-direction accuracy problem. C22 further shows that a correct metric conversion can target one part of that mechanism without uniformly controlling its tail. This negative replacement decision is important evidence: the derivation of γ^2 is valid, but mean γ^2 training alone is not sufficient.

2 Problem and numerical setting

We consider a periodic one-dimensional horizontal domain of length 2π , with free surface $y = \eta(x, t)$, constant depth h , and an irrotational velocity potential ϕ in the fluid. Its trace at the free surface is

$\xi(x, t) = \phi(x, \eta(x, t), t)$. The Dirichlet–Neumann operator is the map from this trace to the non-unit normal derivative,

$$G(\eta; h)\xi = (\phi_y - \eta_x \phi_x)|_{y=\eta}. \quad (1)$$

The definition incorporates the solution of Laplace’s equation in the entire fluid domain. It is therefore the expensive nonlocal component that the surrogate is intended to replace.

For surface elevation η and surface potential ξ , Zakharov’s canonical water-wave equations without surface tension may be written

$$\eta_t = G(\eta)\xi, \quad (2)$$

$$\xi_t = -g\eta - \frac{1}{2}\xi_x^2 + \frac{1}{2} \frac{(G(\eta)\xi + \eta_x \xi_x)^2}{1 + \eta_x^2}. \quad (3)$$

They are Hamilton’s equations for

$$H(\eta, \xi) = \frac{1}{2} \int_0^{2\pi} \xi G(\eta) \xi \, dx + \frac{g}{2} \int_0^{2\pi} \eta^2 \, dx, \quad \eta_t = \frac{\delta H}{\delta \xi}, \quad \xi_t = -\frac{\delta H}{\delta \eta}. \quad (4)$$

This equivalence uses both self-adjointness of $G(\eta)$ in its potential argument and its precise derivative with respect to the surface. Preserving only the former does not suffice.

The production surrogate replaces the expensive DNO G by a learned operator $\mathcal{G}_\theta$, while retaining the geometric formula for ξ_t . Time integration uses a two-stage Gauss–Legendre method with an integrating factor and four fixed-point iterations per step. The DNO model is linear in ξ and built from Craig–Sulem-shaped multiplier sandwiches.

2.1 Reference and surrogate operators

The reference DNO is a truncated Craig–Sulem series evaluated through order six with substantial Fourier zero padding. It is not an exact continuum oracle, but its convergence, self-adjointness, shape-identity residual, and long-rollout conservation are all sufficiently accurate to distinguish the effects discussed below. In particular, the reference shape defect is 10^{-5} or smaller at the failing states, versus $O(1)$ for the learned operator.

The principal baseline is an eight-block Craig–Sulem DNO network. Each block is linear in ξ : a depth-conditioned Fourier multiplier is applied to ξ , multiplied pointwise by learned features of η , and passed through another Fourier multiplier. This is a much stronger inductive bias than a generic FNO and produced excellent one-step validation error. Later checkpoints tied the input and output multipliers so that the learned residual is self-adjoint in ξ . Nothing in the original architecture, however, forces the correct derivative with respect to η .

2.2 The integrating-factor GL2 scheme

Following Xu and Guyenne’s spectral Hamiltonian method [1], split the semi-discrete vector field as

$$u_t = Au + N(u), \quad u = (\eta, \xi)^\top, \quad A = \begin{pmatrix} 0 & G_0 \\ -g & 0 \end{pmatrix}, \quad (5)$$

where $G_0 = |D| \tanh(h|D|)$. With $v(t) = e^{-tA}u(t)$, the stiff linear dispersion is treated exactly and the transformed equation is advanced with the two-stage, fourth-order Gauss–Legendre Runge–Kutta method. Its Butcher coefficients are

$$a_{11} = a_{22} = \frac{1}{4}, \quad a_{12} = \frac{1}{4} - \frac{\sqrt{3}}{6}, \quad a_{21} = \frac{1}{4} + \frac{\sqrt{3}}{6}, \quad b_1 = b_2 = \frac{1}{2}, \quad (6)$$

up to interchange of the two stage labels. The coupled stage equations are solved here with four fixed-point iterations.

Three logically distinct statements must not be conflated:

1. GL2 is a symplectic Runge–Kutta method for a canonical Hamiltonian vector field.
2. A finite number of Picard iterations only approximately solves the implicit stage equations; loss of contractivity can expose a bad state.
3. If the supplied surrogate vector field is not Hamiltonian, exact solution of the GL2 stages still cannot confer the missing Hamiltonian geometry on that field.

The investigation below identifies item 3 as the initiating issue and item 2 as a terminal failure mechanism.

2.3 What counts as success

The practical target is stricter than small one-step validation loss: across the 32-case Tanaka test and the 31 truth-valid Benjamin–Feir cases, the rollout must produce no nonfinite trajectories while retaining competitive median and tail error at final time.

We distinguish four metrics:

One-step DNO error

The relative error in $G(\eta)\xi$ when the model is evaluated on an exact reference state. This measures interpolation on the sampled truth manifold.

NaN rate

The fraction of truth-valid initial conditions for which any surrogate state becomes nonfinite. This is the primary hard stability endpoint.

Divergence rate

The fraction whose final relative state error exceeds the evaluation threshold, including but not limited to nonfinite cases. A finite but physically useless trajectory is not counted as a success.

Final-time error distribution

The median and 95th percentile relative L^2 error in η . Reporting only the median can hide a cap or filter that protects easy cases while producing a catastrophic tail.

The rollout harness stores the integrator state and spectral operations in double precision while leaving the neural network in its trained precision. This matters: an earlier single-precision harness imposed an artificial long-time error floor. Truth-invalid initial conditions are excluded using the reference solver’s own nonfiniteness or excessive Hamiltonian drift, so a surrogate is not penalized for a failed reference trajectory.

3 What the experimental sequence established

Table 2 condenses the main branches recorded in `EXPERIMENTS.md`. The sequence is informative because several apparently plausible interventions improved an intermediate diagnostic without eliminating the failure population.

3.1 Baseline numbers and checkpoint conventions

The principal comparison points are listed in Table 1. “v10” denotes a fresh full-capacity v9 baseline. C1 is a light stage-tangent fine-tune. The first attempted C2 was invalid because a tied-multiplier checkpoint was silently loaded into an untied template; an architecture guard was added, and only the corrected C2 should be interpreted scientifically. C10 is the strongest stage-gain variant and the best pre-C16 NaN count. These distinctions explain why an isolated statement such as “C2 failed” can be misleading without a run path and checkpoint choice.

Model	Tanaka NaN	Tanaka div.	Tanaka med.	BF NaN	BF div.	BF med.
v10 baseline	4/32	5/32	0.0123	2/31	2/31	0.00902
C1 stage match	3/32	3/32	0.0151	0/31	0/31	0.00661
Corrected C2 final	3/32	4/32	0.00907	0/31	0/31	0.00735
C10 final	2/32	3/32	0.0138	0/31	0/31	0.00971

Table 1: Primary pre-C16 comparisons under the double-precision harness. The Benjamin–Feir denominator excludes one truth-invalid case.

For a fine-tune driven by an auxiliary rollout or tangent loss, “best validation” and “best physical checkpoint” need not coincide. The ordinary validation score measures the pointwise supervised objective and can select an early epoch before the auxiliary constraint has acted. This was demonstrated by the corrected C2 run: the final checkpoint had the meaningful regularized behavior, whereas the initially evaluated best-validation checkpoint from the misconfigured run was unusable. For C16, final/latest is therefore the primary rollout checkpoint and best-validation is a secondary control.

3.2 Phase I: establish a trustworthy evaluation

Before asking why a model fails, we removed two ways the evaluation itself could lie.

1. A double-precision rollout harness reduced a spurious long-time error floor by factors of roughly 2.5 on Tanaka and 20 or more on several Benjamin–Feir regimes. An order-six oracle displayed the same single-precision floor, proving it was not neural-model error.
2. Reference trajectories with nonfinite states or Hamiltonian drift above the validity threshold were gated out. For example, one Benjamin–Feir initial condition failed in the truth solver while the surrogate remained finite. Counting it against the surrogate would invert the conclusion.

The residual Tanaka NaNs survive both corrections and are therefore genuine surrogate failures.

3.3 Phase II: test the data-coverage explanation

Additional steep-Tanaka data in v9 improved regimes that genuinely had a coverage gap, including the Tanaka g_1 family, but left 4/32 deep-water Tanaka g_0 NaNs. Random and source-stratified scans found no nonfinite base rows. Earlier sign-flip heuristics were not treated as proofs of corruption; the more decisive test came later: the learned DNO remained highly accurate when evaluated directly on the exact trajectories that would eventually fail under rollout.

Two forms of hard-negative training were then explored. Model-harvested pre-cascade states are circular: the inputs already contain the old model’s spectral contamination, and their approximate targets can inherit it. These packs worsened performance. Clean truth-generated perturbations

avoid that specific flaw and are valid as local learnability probes, but they constrain a finite list of neighborhoods rather than the universal derivative identity. The clean C13 pack trained successfully, but its evaluation was intentionally deprioritized once the structural diagnosis became available. Thus the conclusion is not “additional data can never help”; it is that ordinary pointwise labels are an inefficient and non-general remedy for an operator derivative defect.

3.4 Phase III: test the integrator explanation

The GL2 stage residual becomes large immediately before nonfiniteness, making time-step failure an attractive primary hypothesis. Substep refinements and halving trials, however, moved the failure time only modestly. A matched 2-by-2 comparison—baseline versus C1, with and without Picard-residual containment—left the NaN counts unchanged in every arm. The containment logic was useful operationally because it isolated bad members of a batch, but it did not rescue the affected trajectories.

This phase did reveal a real training signal. C1’s stage-tangent loss cured one of four baseline Tanaka NaNs and both Benjamin–Feir NaNs. Later C10, targeting the production-filtered $k = 64, \dots, 128$ stage gain, reduced Tanaka NaNs to 2/32. Hence sensitivity control is relevant. The failure to reach zero suggested that the indirect stage proxy was not enforcing the defining differential relation strongly or accurately enough.

3.5 Phase IV: suppress the observed spectrum

Hard learned-block cutoffs, soft high-band penalties, run-time high-band limiters, and a checkpoint-carrying output cap all attacked the visible spectral symptom. None reached zero NaNs with acceptable tails. The strongest soft penalty left its own high-band diagnostic essentially unchanged. The C10 run-time limiter retained both Tanaka NaNs and worsened the Benjamin–Feir p95. The C14 hard output projection retained 3/32 Tanaka NaNs, produced six divergent Tanaka cases, and raised Tanaka p95 to 1.25.

The cap’s algebra explains its failure. It imposed

$$\|g_{hi}\| \leq \max(5, 0.1 \|g_{lo}\|). \quad (7)$$

The dangerous $k = 16, \dots, 32$ content belonged to the unconstrained “low” group, growth of that group increased the permitted high-band norm, and the absolute floor was inactive on some failing states. This is a moving relative envelope, not an absolute stability bound.

Finally, ablations showed why editing a trained checkpoint is hazardous. Block 7 appeared individually amplifying, yet zeroing it changed the Tanaka NaN count from 4/32 to 28/32 and the median error by a factor of 62. Its anti- $|k|$ response was compensating the other blocks. Likewise, removing a large derivative feature from an already trained model breaks learned cancellations and does not tell us whether a clean architecture should expose that feature differently.

Two broader lessons emerged. First, deleting an apparently dangerous learned block made the model dramatically worse (28/32 rather than 4/32 Tanaka NaNs), showing that the trained eight-block representation relies on fragile inter-block cancellation. Second, increasing or decreasing parameter count did not track rollout stability. The missing ingredient is a structural identity, not capacity alone.

Experiment family	Main evidence	Conclusion
Clean v9 retraining	Excellent one-step error, yet the baseline retained 4/32 Tanaka NaNs. Additional steep-Tanaka data cured a separate g_1 coverage failure but not the deep-water g_0 failures.	Pointwise coverage is not the principal remaining defect.
GL2 stage regularization	The best corrected fine-tune reduced Tanaka NaNs from 4/32 to 3/32 and Benjamin–Feir NaNs to zero, but did not reach the zero-NaN target and traded easy-case accuracy.	Directionally useful, but an indirect and incompletely matched constraint.
Step residual checks and halving	Picard residual checks detected terminal noncontractivity. Step refinement and attempted halving moved failure times only slightly; per-sample containment did not change the NaN count.	The integrator exposes the failure after the model state is already contaminated.
Hard block cutoffs and output caps	Hard k -cuts and high-band envelopes reduced selected spectral symptoms but produced poor error tails or retained NaNs. The one-block capped model reached 3/32 Tanaka NaNs and severe p95 regression.	Static culling removes physical response and leaves loopholes in the growing mid-band.
Targeted hard negatives	Model-harvested pre-cascade fine-tunes were harmful; clean truth-targeted packs did not provide a general or journal-explainable mechanism.	Do not substitute case selection for an operator identity.
Exact $G_0 + G_1$	A full 32-case CPU rollout gave 0/32 NaNs and 0/32 divergences, but final η error remained too large (median 0.207, p95 0.410).	The analytic null structure is stable, but needs an accurate higher-order residual.

Table 2: Distilled experimental chronology.

4 Frame-by-frame failure anatomy

Let $z_n^* = (\eta_n^*, \xi_n^*)$ be an exact truth state and z_n the corresponding surrogate-rollout state. Separate the model error into the truth-manifold bias and the response induced by rollout displacement:

$$b_n = \mathcal{G}_\theta(z_n^*) - G(z_n^*), \quad (8)$$

$$i_n = \mathcal{G}_\theta(z_n) - \mathcal{G}_\theta(z_n^*), \quad (9)$$

$$q_n = \frac{\|i_n\|_2}{\|z_n - z_n^*\|_2}. \quad (10)$$

The eventual failures remain accurate on truth states: median one-step relative $G\xi$ errors are 1.87×10^{-4} and 1.72×10^{-4} for representative failing cases 5 and 11. The transverse response, however, grows by four orders of magnitude relative to that bias.

Spectrally, the induced response first appears in $k = 32, \dots, 64$, then spreads through $k = 64, \dots, 128$. This precedes the abrupt terminal event. Thus the NaN is not born in one exceptional arithmetic operation; it is the endpoint of a resolved spectral cascade.

The ratio q_n is a secant quantity, not an eigenvalue and not a proof that the full Fréchet derivative has norm 8. It is nevertheless the relevant finite-amplitude response along the error direction actually selected by the rollout. The contrast with healthy cases matters more than its absolute value. In failing case 5 the induced response becomes 2.6×10^4 times the truth-state bias, while a healthy case remains at a gain below one at final time.

The frame order provides a causal discriminator:

1. The pointwise truth-state bias is small and nearly stationary.

Case	Time	State RMS error	Bias RMS	Induced RMS	Secant gain
5	0.8	1.62×10^{-6}	5.12×10^{-6}	5.56×10^{-6}	3.43
5	58.4	2.08×10^{-3}	5.13×10^{-6}	1.24×10^{-2}	5.95
5	66.4	1.67×10^{-2}	5.12×10^{-6}	1.35×10^{-1}	8.11
11	142.4	3.84×10^{-3}	3.15×10^{-6}	2.71×10^{-2}	7.04
11	150.4	9.02×10^{-3}	3.15×10^{-6}	7.59×10^{-2}	8.42
Healthy 0	200.0	5.91×10^{-5}	1.88×10^{-7}	1.96×10^{-5}	0.33
Healthy 6	200.0	3.76×10^{-4}	1.56×10^{-6}	6.71×10^{-4}	1.78

Table 3: Growth of the model response to its own state error.

2. A small state discrepancy develops through ordinary accumulated error.
3. The learned operator maps that discrepancy into a disproportionately large mid-band response.
4. The response broadens spectrally and steepens the state.
5. Only then do base-grid nonlinear products differ materially from their padded counterparts.
6. Finally, the GL2 fixed-point residual ceases to contract and a nonfinite stage is produced.

This ordering rules out “one bad target row directly produces a NaN” and “the first cause is a single aliased product” as descriptions of the observed cases. It does not claim that all possible future NaNs must follow the same path; the conclusion is scoped to the remaining Tanaka population in the audited checkpoints.

5 Hamiltonian inconsistency is the root defect

For a genuine DNO define

$$B = \frac{G(\eta)\xi + \eta_x \xi_x}{1 + \eta_x^2}, \quad V = \xi_x - B\eta_x. \quad (11)$$

The Hadamard shape derivative satisfies, for every perturbation ζ ,

$$D_\eta G(\eta)[\zeta]\xi + G(\eta)(\zeta B) + \partial_x(\zeta V) = 0. \quad (12)$$

This is an identity of operators applied to ξ , not merely an energy conservation statement along one trajectory. Its importance can be seen by a short variational calculation. Let

$$K_\theta(\eta, \xi) = \frac{1}{2} \langle \xi, \mathcal{G}_\theta(\eta)\xi \rangle, \quad (13)$$

where $\langle f, g \rangle = \int_0^{2\pi} fg \, dx$. Assuming self-adjointness of $\mathcal{G}_\theta(\eta)$ and Eq. (12), periodic integration by parts gives

$$D_\eta K_\theta[\zeta] = \frac{1}{2} \langle \xi, D_\eta \mathcal{G}_\theta(\eta)[\zeta]\xi \rangle \quad (14)$$

$$= -\frac{1}{2} \langle \xi, \mathcal{G}_\theta(\eta)(\zeta B_\theta) \rangle - \frac{1}{2} \langle \xi, \partial_x(\zeta V_\theta) \rangle \quad (15)$$

$$= -\frac{1}{2} \langle \mathcal{G}_\theta(\eta)\xi, \zeta B_\theta \rangle + \frac{1}{2} \langle \xi_x, \zeta V_\theta \rangle \quad (16)$$

$$= \frac{1}{2} \int_0^{2\pi} \zeta (V_\theta \xi_x - B_\theta \mathcal{G}_\theta(\eta) \xi) dx. \quad (17)$$

Therefore

$$-\frac{\delta K_\theta}{\delta \eta} = \frac{1}{2} (B_\theta \mathcal{G}_\theta(\eta) \xi - V_\theta \xi_x). \quad (18)$$

Using $V_\theta = \xi_x - B_\theta \eta_x$ and $B_\theta = (\mathcal{G}_\theta \xi + \eta_x \xi_x)/(1 + \eta_x^2)$, the right-hand side becomes

$$-\frac{1}{2} \xi_x^2 + \frac{1}{2} \frac{(\mathcal{G}_\theta(\eta) \xi + \eta_x \xi_x)^2}{1 + \eta_x^2}, \quad (19)$$

which is exactly the nonlinear kinetic contribution used in Eq. (3). This proves the claimed equivalence.

Conversely, tying the learned Fourier multipliers makes $\mathcal{G}_\theta$ self-adjoint and guarantees $\delta K_\theta / \delta \xi = \mathcal{G}_\theta \xi$, but it says nothing about $\delta K_\theta / \delta \eta$. The surface derivative must separately obey Eq. (12). This is the precise sense in which self-adjointness is necessary but not sufficient.

5.1 The measured defect

For diagnosis, the continuum derivative is evaluated by automatic differentiation of the model energy and compared with the geometric expression on the same discrete grid. Both sides undergo the production $|k| \leq 128$ projection. Schematically, the reported relative defect is

$$d_{\text{shape}} = \frac{\|P_{\leq 128}(\xi_{t,\text{geom}} + \delta_\eta K_\theta)\|_{H^1}}{\|P_{\leq 128} \xi_{t,\text{geom}}\|_{H^1} + \varepsilon}. \quad (20)$$

The same normalization, projection, and differentiation pipeline is used for the order-six reference control.

The measured discrepancy, after the production projection to $|k| \leq 128$, is summarized below. The exact order-six DNO passes the same discrete test, ruling out a diagnostic normalization error.

Case/time	Value error	Defect at truth	Defect at rollout	Order-six defect
Healthy 0 / 200.0	6.47×10^{-5}	0.0896	0.2966	2.0×10^{-9}
5 / 33.6	2.34×10^{-4}	1.394	4.975	1.43×10^{-5}
11 / 189.6	2.15×10^{-4}	0.502	1.117	3.43×10^{-6}
22 / 45.6	5.54×10^{-4}	2.724	7.481	3.82×10^{-6}

Table 4: Small pointwise error coexisting with a large shape-derivative defect.

This resolves an apparent contradiction. Gauss–Legendre methods are symplectic for the Hamiltonian vector field supplied to them. The surrogate field is not, in general, the gradient of one Hamiltonian because its η_t and ξ_t components use incompatible derivatives of $\mathcal{G}_\theta$. The time integrator therefore cannot protect the rollout from the learned geometric inconsistency.

One can also phrase the problem through the error equation. Write the truth and surrogate vector fields as F and F_θ , and let $e = z_\theta - z$. Before the error is large,

$$e_t = \underbrace{F_\theta(z) - F(z)}_{\text{truth-manifold bias}} + \underbrace{DF_\theta(z)e}_{\text{propagation by learned tangent map}} + O(\|e\|^2). \quad (21)$$

Supervised one-step fitting directly attacks the first term. The aligned NPZ analysis shows that this term stays small while the second becomes dominant. The Hadamard defect identifies a specific

incorrect component of DF_θ , rather than appealing vaguely to “error compounding.” The spectral cascade is error compounding, but it is error compounding through a geometrically wrong tangent operator.

6 The missing spectral null form

The exact first Craig–Sulem correction is

$$G_1(\eta)\xi = -G_0(\eta G_0\xi) - \partial_x(\eta \partial_x \xi). \quad (22)$$

Its Fourier interaction kernel is

$$\widehat{G_1(\eta)\xi}_k = \sum_{k'} \underbrace{(kk' - g_0(k)g_0(k'))}_{\mathcal{K}_1(k,k';h)} \widehat{\eta}_{k-k'} \widehat{\xi}_{k'}, \quad g_0(k) = |k| \tanh(h|k|). \quad (23)$$

The sign is worth checking. The first term in Eq. (22) contributes $-g_0(k)g_0(k')$, while $-\partial_x(\eta \partial_x \xi)$ contributes $+kk'$. In deep water, $g_0(k) = |k|$, so $\mathcal{K}_1 = 0$ whenever k and k' have the same sign. This is not a learned small coefficient: it is exact cancellation between two individually large derivative terms. At finite depth the cancellation is exponentially accurate when $h|k|$ and $h|k'|$ are large, as they are near the observed modes.

The null form is physically important in the failing Tanaka regime. A transfer-kernel probe at a truth surface found that the learned model maps $k_{\text{in}} = 117$ to $k_{\text{out}} = 115$ with amplitude 985.67, whereas the order-six reference amplitude is 1.47×10^{-5} , a ratio of approximately 6.7×10^7 . The flat diagonal response at this mode is about 5.9×10^4 , so 985.67 is only about 1.7% of the diagonal amplitude. That explains why an aggregate pointwise error metric can appear excellent while a repeatedly injected off-diagonal sideband matters dynamically. The learned model can represent the cancellation but does not enforce it.

Across all sampled Tanaka truth states, replacing G_0 by exact $G_0 + G_1$ reduces median relative $G\xi$ error from 0.1496 to 0.00555 and p95 from 0.2089 to 0.01373. More importantly, it makes the high-frequency null form exact rather than dependent on cancellation among learned blocks.

The analytic-only rollout is a particularly discriminating experiment. Exact $G_0 + G_1$, with no cap and no learned residual, yields 0/32 NaNs and 0/32 divergences. It nevertheless gives final η median 0.2071 and p95 0.4096. Hence two statements are simultaneously true:

1. the first-order null structure removes the catastrophic mechanism in this test; and
2. neglected $G_2, G_3, \dots$ terms are essential for useful long-time phase and amplitude accuracy.

This is why the proposed solution is not to abandon learning, but to prevent the learned residual from overwriting the analytically known first-order term.

7 Aliasing and the role of the integrator

For truncated spectra, a physical-space product computes a circular convolution. If f and g contain modes near the retained cutoff, terms with true wavenumber outside the grid wrap into lower modes. The nonlinear Zakharov right-hand side contains products and a rational expression in η_x , ξ_x , and $G\xi$, so this mechanism becomes severe once the surrogate has populated its upper retained bands.

Xu and Guyenne explicitly identify aliasing in both the DNO expansion and the equations of motion, and extend the spectra by a factor of two for nonlinear terms in the equations of motion [1,

Sec. 3.3]. Our order-six reference DNO already uses padding, but the original surrogate and truth rollout paths evaluated the quadratic Zakharov products at base resolution. The correction now implemented is:

1. Fourier-interpolate η_x , ξ_x , and $G\xi$ onto a grid with twice as many points;
2. evaluate the nonlinear expression pointwise on that padded grid; and
3. truncate the result back to the production spectrum before applying the usual retained-mode projection.

Two-times Fourier padding is therefore a necessary discretization correction. On healthy trajectories, base-grid and padded right-hand sides agree to about 10^{-11} . Near the last finite frames, the discrepancy becomes large:

Case	Last finite time	Relative RHS difference	First $> 10^{-3}$	First > 0.1
Healthy 0	200.0	2.71×10^{-11}	never	never
Healthy 6	200.0	1.78×10^{-11}	never	never
5	33.6	0.753	32.0	33.6
11	189.6	0.819	187.2	188.8
22	45.6	0.0829	45.6	never

Table 5: Difference between base-grid and two-times-padded Zakharov right-hand sides.

The large terminal discrepancy alone does not establish that aliasing started the cascade. A causal replay does: restarting each failing state with padded products delays the first nonfinite value by only 0.05, 0.08, and 0.05 time units for cases 5, 11, and 22. A healthy replay agrees to 3.5×10^{-15} . Aliasing is therefore a selective terminal amplifier, not the initiator in these cases. The padding fix should be retained regardless of C16’s outcome because it restores the intended pseudospectral discretization without perturbing resolved healthy trajectories.

Likewise, more Picard work or smaller steps cannot undo the preceding operator-induced drift. A residual check remains valuable as a reported fail-safe, and adaptive halving may rescue milder future failures, but it is not a substitute for restoring Hamiltonian compatibility.

There is no conflict with the well-known long-time advantages of symplectic integration. Those advantages concern the discretization of a Hamiltonian vector field and usually imply bounded, structured energy error over long times, rather than exact conservation at each step. Here the learned vector field fails the compatibility calculation in Eq. (17). Moreover, the practical scheme uses a fixed number of nonlinear iterations, so once the state enters a high-gain region it may not even realize the exact implicit GL2 map. The integrator is important, but it is downstream of the operator geometry.

8 C16: the resulting model and objective

The proposed clean retraining uses

$$\mathcal{G}_\theta(\eta) = G_0 + G_1(\eta) + \mathcal{R}_\theta(\eta), \quad \mathcal{R}_\theta(\eta) = O(\eta^2). \quad (24)$$

The analytic G_1 path is uncut and evaluated with double-precision spectral arithmetic. The learned residual uses eight tied, self-adjoint blocks with 256 branches each. A bias-free feature trunk followed by $f \mapsto f \tanh(f)$ ensures

$$\mathcal{R}_\theta(0)\xi = 0, \quad D_\eta \mathcal{R}_\theta(0)[\zeta]\xi = 0 \quad (25)$$

for every parameter value. The residual sum is scaled by $1/\sqrt{8 \cdot 256}$, making the size of a fresh Adam update independent of the chosen parallel rank without changing the function class.

8.1 Residual parameterization and its invariants

Suppressing depth and normalization factors, the residual has the schematic form

$$\mathcal{R}_\theta(\eta)\xi = \frac{1}{\sqrt{N_b N_r}} \sum_{b=1}^{N_b} \sum_{r=1}^{N_r} M_{b,r}(D, h) [\Phi_{b,r}(\eta) M_{b,r}(D, h)\xi], \quad N_b = 8, \quad N_r = 256. \quad (26)$$

The same real Fourier multiplier appears on both sides of the pointwise multiplication. For fixed η , each term is self-adjoint:

$$\langle \psi, M[\Phi M\xi] \rangle = \langle M\psi, \Phi M\xi \rangle = \langle \Phi M\psi, M\xi \rangle = \langle M[\Phi M\psi], \xi \rangle. \quad (27)$$

The proof uses self-adjointness of the real multiplier and of multiplication by the real function Φ . Summation and positive scalar normalization preserve the property.

For the order statement, let the bias-free eta trunk produce $f(\eta) = L\eta + O(\|\eta\|^2)$ near the flat surface. Since $\tanh f = f + O(f^3)$,

$$f(\eta) \tanh f(\eta) = (L\eta)^2 + O(\|\eta\|^3). \quad (28)$$

All subsequent eta projections are bias-free. Consequently every $\Phi_{b,r}(0) = 0$ and $D_\eta \Phi_{b,r}(0) = 0$, proving the two structural zeros. The bounded-growth lift was chosen instead of raw f^2 : it retains quadratic behavior near zero but grows only linearly in $|f|$ when derivative features are large. Learned mixing before the lift still permits general quadratic cross-interactions; this is not a restriction to literal squares of the physical input channels.

The architecture contains 860,928 trainable parameters. It retains the full eight-block capacity because the one-block and block-deletion experiments did not support a capacity reduction. There is no hard spectral cutoff, no output cap, no PSD hinge, and no Tanaka-specific data pack in C16. Thus a successful result can be attributed to the null-structured backbone and direct shape constraint rather than to spectral culling.

8.2 Finite-secant Hadamard loss

The randomized finite-secant residual is

$$S_\theta = \frac{\mathcal{G}_\theta(\eta + \epsilon\zeta)\xi - \mathcal{G}_\theta(\eta)\xi}{\epsilon} + \mathcal{G}_\theta(\eta)(\zeta B_\theta) + \partial_x(\zeta V_\theta), \quad (29)$$

and the training objective is

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{shape}} \mathbb{E}_\zeta \frac{\|WP_{\leq 128} S_\theta\|_2^2}{\|WP_{\leq 128} (\mathcal{G}_\theta(\eta)(\zeta B_\theta) + \partial_x(\zeta V_\theta))\|_2^2 + \delta}. \quad (30)$$

Here W is the supervised Sobolev weight. Relative, zero-mean, inverse-Sobolev probes cover low and mid/high modes without selecting specific Tanaka cases. The regularizer is evaluated in double precision on a global microbatch of eight every 16 optimizer steps, with $\lambda_{\text{shape}} = 10^{-2}$ introduced through a warmup.

More explicitly, a random Fourier probe is projected to zero mean and scaled by an inverse Sobolev symbol so its high modes do not dominate merely because derivatives multiply by $|k|$.

Its physical RMS is chosen relative to the RMS of the current η , subject to a small floor, and $\epsilon \in [10^{-3}, 3 \times 10^{-3}]$ is sampled as a relative secant size. The two model evaluations use exactly the production input normalization, output de-normalization, filtering, and output-mean subtraction. Both the secant residual and its normalizing forcing are measured after projection to $|k| \leq 128$ in the same Sobolev metric used by supervised training.

The denominator in Eq. (30) is essential. An absolute residual would overweight large-amplitude samples and permit the loss scale to change with probe amplitude. The relative form asks whether the identity error is small compared with the physical terms that should cancel. The floor δ only regularizes nearly trivial forcing. A finite secant, rather than an exact Jacobian norm, avoids differentiating through a full Jacobian and keeps the cost compatible with sparse microbatch evaluation.

This loss is related to, but cleaner than, the earlier GL2 stage regularizer. The stage loss asks whether one particular discretized short-time map behaves well under selected perturbations; it mixes DNO error, time-discretization details, filters, and stage construction. The Hadamard loss asks directly for the continuum operator identity that makes the Zakharov vector field Hamiltonian. It is therefore cheaper to interpret and easier to defend in a mathematical presentation.

8.3 Optimizer calibration

The first implementation exposed a separate optimizer parameterization issue: hundreds of thousands of zero-initialized projection weights received aligned Adam sign updates and were summed without rank normalization. Before the normalization, a single real-state step at learning rate 2×10^{-5} changed loss from 0.02463 to 0.3201. After normalization, the same step reduced it to 0.02174. A 1% v9 GPU smoke then completed with train loss 0.01361 and validation loss 0.01311, with no surge or nonfinite metric.

The mechanism of the initial optimizer instability deserves emphasis because it is independent of the NaN-rollout diagnosis. At initialization, the analytic baseline already has small loss and the per-block eta-to-branch projection kernels are zero. Adam normalizes each nonzero coordinate by its own second moment, so the first update is approximately a sign step in each of 524,288 projection coordinates. All coordinates are trained against the same residual and their function-space effects add coherently across eight blocks and 256 branches. A parameter-space learning rate that was harmless in a warm-started model therefore produced an enormous first function-space step in the fresh model.

The factor $1/\sqrt{N_b N_r} = 1/\sqrt{2048}$ is a fan-in normalization for the parallel residual paths. It preserves exact $G_0 + G_1$ at initialization, the order-two zero, self-adjointness, parameter count, and the set of functions representable after rescaling downstream weights. CPU calibration gave

Parameterization	Loss before step	Loss after AdamW step at 2×10^{-5}
Unnormalized order-two residual	0.024634	0.320115
Fan-in-normalized residual	0.024634	0.021744

The data-gradient to Hadamard-gradient norm ratio was 0.0662, and the chosen Hadamard weight 0.01 makes the auxiliary gradient influential but not dominant at initialization. Both the learning rate and Hadamard weight are warmed up over 500 steps.

8.4 What C16 guarantees, and what it does not

For every value of its trained parameters, C16 guarantees:

- exact $G_0 + G_1$ behavior through first order at $\eta = 0$;
- linearity and self-adjointness in ξ for the learned residual;
- no learned $O(1)$ or $O(\eta)$ term that can overwrite the analytic first-order null form; and
- a consistently two-times-dealiased Zakharov nonlinear right-hand side.

It does *not* guarantee the full nonlinear Hadamard identity for finite η ; that is encouraged statistically by Eq. (30). Nor does it prove bounded rollouts, exact energy conservation, or uniform operator error outside the sampled distribution. Calling it an “explicit bounded residual architecture” would therefore be inaccurate. Its principle is geometric compatibility and asymptotic null structure, not a hard global norm bound.

9 C16 launch-time acceptance criteria (historical)

At the time of the original 9 July report, a budget-capped C16 retraining was running on two H100 GPUs in the personal Modal workspace. This section retains the pre-specified diagnostics and gates against which that structural test was judged; the completed C16–C22 results are reported in Section 10. The launch used clean v9 data, batch size 1024, learning rate 2×10^{-5} , 500-step learning-rate and Hadamard warmups, and 16 epochs. The abbreviated run was intended to obtain the strongest checkpoint possible within approximately 30 dollars of workspace credit; epoch 16 was not assumed to be fully converged.

The first two completed epochs were healthy:

Epoch	Train loss	Validation loss	Hadamard loss	Defect RMS
1	0.009919	0.008740	0.04381	0.1700
2	0.008472	0.008376	0.03104	0.1482

The regularizer was active on 373 of 5964 batches in each epoch, matching the intended frequency 1/16. Its mean effective weight reached 9.57×10^{-3} in epoch 1 and the full 10^{-2} in epoch 2. These were encouraging training diagnostics, not rollout evidence. In particular, the random-probe defect RMS could not yet be compared directly with the case-specific defect table without running the same held-state diagnostic.

An earlier Modal attempt had stopped one third of the way through epoch 1 because the local monitoring client was explicitly interrupted; Modal logs show a cancellation signal rather than a Python, JAX, or model exception. No checkpoint was produced by that attempt. The subsequent 16-epoch run was a fresh relaunch and was judged independently.

The structural hypothesis was to be accepted only if rollout evaluation showed all of the following:

1. zero Tanaka NaNs and divergences among 32 truth-valid cases;
2. zero scored Benjamin–Feir NaNs and divergences among 31 cases;
3. no material median or p95 regression relative to the strongest prior checkpoints;
4. at least a two-order-of-magnitude reduction in the failing-state shape defect;
5. no pre-cascade secant-gain growth toward 7–8; and
6. healthy GL2 fixed-point residuals without relying on output caps, hard spectral cutoffs, or case-specific damping.

The main scientific conclusion does not depend on the outcome of one training run: long-rollout stability requires approximating the geometry of the DNO in addition to its values. C16 is the first experiment in this sequence that directly enforces the missing first-order null form and shape derivative while retaining the capacity needed for higher-order accuracy.

9.1 Pre-registered post-training decision sequence

To avoid another ambiguous experiment, evaluation was to proceed in the following order:

1. Verify checkpoint/config integrity and run the held-state Hadamard defect table on the final checkpoint. This is cheap and directly tests whether the intended loss changed the diagnosed quantity.
2. Evaluate the final checkpoint on Tanaka g_0 and Benjamin–Feir g_1 with cached truth, the double-precision harness, batched surrogate execution, and no run-time limiter or damping.
3. If final is negative, evaluate best-validation only as a checkpoint selection control; do not reinterpret it as a separate architecture.
4. For every remaining failure, regenerate the frame-aligned secant, spectral-band, RHS-alias, and GL2-residual traces. The question is not merely whether the failure count moved, but whether the causal signature changed.
5. Only after the two decisive regimes pass should the full multi-regime suite be run to rule out regressions in shallow water, random seas, Stokes waves, and modal instability.

10 Completed C16–C22 results and the remaining phase tail

The completed sequence separates two questions that were conflated when NaNs were still present. The first is whether the learned vector field excites the transverse mid/high-frequency instability described above. The second is whether a stable learned vector field advances coherent waves at sufficiently accurate phase speeds over $T = 200$. C16 and its descendants resolve the first question on the registered suite; C21 gives the best aggregate answer to the second so far, but leaves a measurable differential-phase tail. C22 tests the derived phase-sensitivity normalization directly and improves the targeted rigid tail without satisfying the broader replacement criteria.

10.1 From the structural C16 result to the C20 control

The clean C16 epoch-39 checkpoint was evaluated on 32 initial conditions from each of ten families. Every one of the 320 predicted trajectories remained finite. The only counted final divergences were one finite phase-error case in each Tanaka family; all eight other families had zero. The Tanaka g_0/g_1 median and 95th-percentile final-elevation errors were 0.08643/0.51111 and 0.11541/0.66002, respectively. Thus the $G_0 + G_1 + O(\eta^2)$, self-adjoint, Hadamard-trained construction passed the primary stability test but did not yet provide the desired phase accuracy.

C20 retained this structural architecture and clean v9 training distribution, and added the accepted mode-balanced objective, the corrected localized translation-tangent term at coefficient 10, and Hadamard coefficient 0.01 in a fresh 40-epoch run. It also produced 0/320 nonfinite trajectories. On the 305 cases whose reference trajectories passed the pre-existing truth-drift test, C20’s median, 95th percentile, and mean errors were 0.007201, 0.16082, and 0.03464. It is therefore the clean control for asking whether a change in the phase loss itself, rather than a new architecture or data distribution, improves the remaining tail.

10.2 C21 intervention and optimization evidence

C21 is a deliberately narrow diagnostic continuation from C20, not a new architecture or a fresh full training run. Adam was reset and one full v9 epoch was run at learning rate 2×10^{-6} , with the localized-tangent coefficient increased from 10 to 100. Batch size 1024, mode-balanced coefficient 6, Hadamard coefficient 0.01, model parameters, data, and all other losses were unchanged. The resulting train/composite-validation/H1-validation/ modal-validation losses were 0.00694792/0.00684961/0.00664756/ 3.36764×10^{-5} . The expected and observed Hadamard evaluations agreed exactly at 373/373.

The larger coefficient acted in the intended direction without dominating the fit:

Diagnostic	C20	C21	Relative change
H1 validation loss	6.65323×10^{-3}	6.64756×10^{-3}	−0.085%
Modal validation loss	3.24949×10^{-5}	3.36764×10^{-5}	+3.64%
Raw localized-tangent loss	5.7188×10^{-7}	4.7457×10^{-7}	−17.0%
Mean relative local-speed error	2.4925×10^{-4}	2.2980×10^{-4}	−7.8%
Weighted tangent fraction of train loss	0.0829%	0.683%	—

These numbers establish that the intervention changed its target while remaining subdominant. They do not by themselves establish a rollout improvement; that conclusion comes from the fixed-suite evaluation below.

10.3 Full-suite performance

Table 6 reports final relative elevation error under the same cached-truth, double-precision-harness protocol. The number in the second column is the count retained after applying the reference-only Hamiltonian drift validity mask. That mask excludes 15 reference trajectories; it is not a surrogate-finiteness filter.

Family	Valid	C20 median	C20 p95	C21 median	C21 p95
Tanaka g_0	32	0.022810	0.298468	0.021041	0.260258
Tanaka g_1	32	0.052117	0.403609	0.034384	0.382378
Benjamin–Feir g_0	32	0.007292	0.149715	0.007240	0.128859
Benjamin–Feir g_1	31	0.016128	0.072102	0.016341	0.070135
Benjamin–Feir modal	31	0.011504	0.108985	0.012146	0.114576
Shallow nonlinear	20	0.016757	0.164532	0.016610	0.163205
Random sea, deep	32	0.001449	0.013537	0.001495	0.013467
Random sea, finite depth	32	0.002513	0.024598	0.002554	0.023935
Stokes, deep	32	0.000462	0.001323	0.000459	0.001314
Stokes, finite depth	31	0.003858	0.033379	0.003297	0.036508

Table 6: Matched C20 and C21 final-elevation errors on the fixed ten-family suite. Medians and p95 values use only truth-valid cases, while the separate finiteness audit scans every predicted array.

Across the eight non-Tanaka families, comprising 241 truth-valid cases, the C20-to-C21 median/p95/mean change is

$$0.005366/0.07840/0.01803 \longrightarrow 0.005424/0.07490/0.01785,$$

or +1.08%/ − 4.47%/ − 0.98%. The corresponding counts above final-error thresholds improve as follows:

$$(N_{>0.25}, N_{>0.50}, N_{>0.75}, N_{>1.00}) : \quad 3/0/0/0 \longrightarrow 2/0/0/0.$$

The Benjamin–Feir g_0 case 23 tail changes only from 0.36379 to 0.37364 and remains below its pre-specified 0.40 gate.

Combining those cases with the 64 Tanaka trajectories yields 305 truth-valid cases. The full-suite median/p95/mean improves

$$0.007201/0.16082/0.03464 \quad \longrightarrow \quad 0.006866/0.15369/0.03205,$$

or $-4.66\%/ -4.43\%/ -7.47\%$. Counts above the same four thresholds improve from 10/1/1/1 to 8/1/0/0. An explicit array scan found every C21 η , ξ , and $G\xi$ value finite in all 320 trajectories, including the 15 with truth-invalid references, and every family has zero counted final divergences. These results promote C21 as the best aggregate accuracy/stability checkpoint. C20 is nevertheless retained as the clean phase-loss control because C21 changes the old loss weight and exhibits the tradeoff below.

10.4 The decisive tradeoff: rigid versus differential phase

The strongest C20 outlier, Tanaka g_0 case 27, is a narrow shallow solitary wave whose error was almost a rigid translation. C21 reduces its raw/aligned final error from 1.12696/0.05061 to 0.33642/0.03050, reduces the best-fit shift from 7.45 to 1.69 grid points, and changes final learned-Hamiltonian drift from -5.43% to -2.84% . This is direct evidence that stronger tangent supervision can correct a coherent speed bias.

The cost is Tanaka g_1 case 24, a two-crest state. Its raw/aligned error changes as

$$0.45866/0.14579 \quad \longrightarrow \quad 0.65909/0.24546,$$

and the g_1 aligned p95 correspondingly rises from 0.09723 to 0.14165. At final time its two crests are displaced by $+6.21$ and -0.24 grid points, yet after aligning the crests separately their local shape errors are only 3.1% and 0.2%. The displacement split grows smoothly over the rollout. It is therefore a differential local-speed allocation error, not a sudden shape failure.

More generally, C21’s six Tanaka cases above raw error 0.25 comprise four nearly rigid translations and two differential multi-crest translations. Across these cases the final high-band energy is 0.83–1.07 times the truth, the maximum-slope ratio is 0.93–1.02, learned-Hamiltonian drift remains within approximately $\pm 4.4\%$, and every saved field is finite. These facts rule out recurrence of the former mid/high-band cascade and provide no signature of an abrupt GL2 failure. Moreover, translation invariance of the Hamiltonian means that accurate energy alone cannot determine the point occupied on a translation group orbit. The remaining tail is a secular phase-accuracy problem.

10.5 From local speed error to rollout error

Let $T_s\eta(x) = \eta(x - s)$. For a small displacement s ,

$$\frac{\|T_s\eta - \eta\|_2}{\|\eta\|_2} = |s| \kappa(\eta) + O(s^2), \quad \kappa(\eta) = \frac{\|\eta_x\|_2}{\|\eta\|_2}. \quad (31)$$

If $\delta c(t)$ is a coherent speed defect, then

$$e_\eta(T) \simeq \kappa(\eta(T)) \left| \int_0^T \delta c(t) dt \right|. \quad (32)$$

Thus a small one-step DNO error can produce an order-one final error without exponential growth, and a narrow profile is less tolerant of the same speed error than a broad one.

The implemented tangent regularizer already estimates a local speed defect. Let $e_q = q_\theta - q_*$, where $q = G(\eta; h)\xi$, and let K_h be the periodic Gaussian with multiplier $\exp[-(hk)^2/2]$. Define

$$\begin{aligned} A_h &= K_h * (\eta_x e_q), & B_h &= \max\{K_h * (\eta_x^2), 0\}, \\ \delta c_h &= -\frac{A_h}{B_h + 10^{-3} \max_x B_h + 10^{-30}}, & d\mu_h &= \frac{B_h dx}{\int B_h dx}. \end{aligned} \quad (33)$$

The current per-state loss is

$$\ell_{\text{tan}} = \int \left(\frac{\delta c_h}{\sqrt{gh}} \right)^2 d\mu_h. \quad (34)$$

This is a meaningful dimensionless fractional-speed error, but it is not the instantaneous growth of the reported relative-elevation error. A normalized Gaussian preserves the integral, so $\int B_h dx = \int \eta_x^2 dx$, up to the inactive roundoff clamp in Eq. (33). Consequently

$$\boxed{\gamma^2 := \frac{\int B_h(x) \delta c_h(x)^2 dx}{\int \eta(x)^2 dx} = gh \kappa(\eta)^2 \ell_{\text{tan}}.} \quad (35)$$

When the tangent component of the DNO defect is $e_{\eta,t} \simeq -\delta c_h \eta_x$, Eq. (35) is the Gaussian-local estimate of $\|e_{\eta,t}\|_2^2 / \|\eta\|_2^2$, the squared instantaneous growth rate of the actual relative-error metric. The equality in Eq. (35) is exact for the stated continuous normalized Gaussian; the implementation should replace the elevation denominator by $\int \eta^2 dx + \varepsilon_\eta$ only for numerical safety and record the resulting regularization factor. A common horizon factor T^2 can be absorbed into the loss coefficient. The essential correction is the sample-dependent sensitivity $gh\kappa^2$, not another universal increase of the coefficient multiplying Eq. (34).

The fixed 64-state Tanaka audit supports this conversion, but the frame-zero and final-profile quantities must be distinguished. On the C20 initial truth states, the minimum, median, 90th percentile, and maximum κ values are 0.4819860400, 6.3151146892, 19.3302364980, 35.2952751152, with Pearson $\text{corr}(\log h, \log \kappa) = -0.9706889755$. The corresponding minimum, median, and maximum $gh\kappa^2$ factors are 0.06240194586, 2.028865511, 14.45894588. The final truth profiles instead have κ statistics 0.4075981345, 6.318302638, 20.50914397, 33.33393313 and Pearson log-log correlation -0.9620172732 . Case 27's final-profile $\kappa = 33.33393313$ is the value relevant to its shift tolerance: $0.25/\kappa = 0.00750$, while exact periodic Fourier translation and bisection give 0.007566.

At C20 frame zero, the Pearson correlation of log score with log final raw elevation error is 0.7369574728 for γ^2 , compared with 0.6502210258 for ℓ_{tan} . This improvement is specific to logarithmic tail ordering: the raw-scale Pearson correlations are 0.794757 and 0.796616, respectively, so the raw linear correlation does not improve. Case 27 ranks first and case 24 seventh under the derived factor. These are diagnostic correlations rather than a training result, but they show that the omitted factor has the correct scale and logarithmic case ordering.

For multi-crest states, no discrete crest identities are required. With $\overline{\delta c} = \int \delta c_h d\mu_h$, write

$$\begin{aligned} \gamma_{\text{rigid}}^2 &= \kappa^2 \overline{\delta c}^2, \\ \gamma_{\text{diff}}^2 &= \kappa^2 \int (\delta c_h - \overline{\delta c})^2 d\mu_h, & \gamma^2 &= \gamma_{\text{rigid}}^2 + \gamma_{\text{diff}}^2. \end{aligned} \quad (36)$$

The first term is the squared energy-weighted spatial-mean contribution of the local speed defect; the second is its spatial variance contribution. The latter can detect different local speeds such as case 24 and remains well-defined when crests approach or exchange order. It is not, however, mathematically equivalent to inter-crest variance: nonuniformity within one narrow crest also enters the second term. The frame-zero discriminator below shows that this distinction is operationally important.

10.6 C22: controlled mean phase-growth ablation

C22 executed the pre-registered first ablation. It began from the same C20 epoch-40 best checkpoint used to initialize C21, thereby isolating the loss normalization from C21’s tenfold increase of the old coefficient. Adam was reset; one full v9 epoch used batch size 1024, learning rate 2×10^{-6} with a 200-step warmup, mode-balanced coefficient 6, and Hadamard coefficient 0.01. The old tangent coefficient was set to zero and the full-v9-calibrated phase-growth coefficient was 10. The architecture, data, and remaining objective terms were unchanged. Unlike the old tangent term in C20, the phase-growth term was included in validation and checkpoint selection.

Training completed without a numerical or accounting anomaly. The exact epoch metrics are:

Metric	C22 value
Train loss	0.0069397871
Composite validation loss	0.0068893890
H1 validation loss	0.0066473825
Modal validation loss	3.3678880×10^{-5}
Phase-growth validation loss	3.9933183×10^{-6}
Phase-growth train loss	4.0587997×10^{-6}
Rigid phase-growth component	2.4040945×10^{-7}
Differential phase-growth component	3.8183903×10^{-6}

The two phase components sum to the reported train phase loss within logging precision. After multiplication by coefficient 10, the phase term contributes 0.585% of the train objective, inside the intended 0.5–1.0% range. The Hadamard schedule fired on exactly 373/373 expected batches, and modal clipping was zero. Thus a negative rollout decision cannot be attributed to an absent auxiliary, coefficient-scale error, missed Hadamard schedule, or clipping.

The matched 32+32 Tanaka evaluation used the same cached truth and continuous periodic alignment as C20 and C21. Table 7 gives the family-level raw and aligned statistics.

Family	Statistic	C20	C21	C22
Tanaka g_0	Raw median	0.02280974	0.02104052	0.03370219
	Raw p95	0.29846847	0.26025790	0.20827000
	Aligned median	0.00517506	0.00477889	0.00536654
	Aligned p95	0.04630348	0.03501889	0.03285937
Tanaka g_1	Raw median	0.05211732	0.03438386	0.04461333
	Raw p95	0.40360892	0.38237807	0.40356176
	Aligned median	0.00909044	0.00903635	0.01061485
	Aligned p95	0.09723028	0.14164622	0.13773014

Table 7: Matched Tanaka final-elevation errors for the C20 control, promoted C21 checkpoint, and C22 mean- γ^2 ablation.

The combined distributions make the body–tail tradeoff explicit:

Run	Raw median	Raw p95	Aligned median	Aligned p95
C20	0.03303033	0.35831433	0.00668897	0.05714684
C21	0.02788697	0.34413996	0.00583046	0.05621389
C22	0.03444904	0.37679914	0.00834420	0.05643572

Strict raw counts above 0.25/0.50/0.75/1.00 are, respectively, 7/1/1/1, 6/1/0/0, and 6/1/0/0 for C20, C21, and C22. The aligned counts are 1/0/0/0 for all three. Every C22 predicted field is finite in both families, giving 0/64 nonfinite trajectories.

The named cases show why the intervention is a partial mechanistic success but not a replacement:

Run	Case 27 raw	Case 27 aligned	Case 24 raw	Case 24 aligned
C20	1.12695932	0.05061293	0.45866336	0.14579412
C21	0.33642204	0.03049814	0.65908811	0.24545740
C22	0.18046794	0.02688441	0.63952219	0.23661902

The frame audit resolves the remaining differential tail more sharply. Table 8 shows smooth, secular growth rather than a terminal jump. At final time, case 24’s two crest shifts are [5.984, −0.224]

Case	$t = 50$	$t = 100$	$t = 150$	$t = 200$
Tanaka g_1 case 24	0.0250/0.0140	0.1671/0.0656	0.3810/0.1413	0.6395/0.2366
Tanaka g_1 case 22	0.0201/0.0200	0.0903/0.0798	0.2243/0.1933	0.4158/0.3617

Table 8: C22 raw/aligned relative elevation error at four saved times.

grid points and its independently aligned local errors are [2.93%, 0.19%]. Case 22’s three crest shifts are [4.255, −2.001, 0.435], with local errors [4.05%, 1.69%, 1.18%]. Thus both predictions retain accurate local crest shapes while allocating different speed errors to different crests. Case 27 is complementary: its C22 framewise repair is bounded and nonmonotone, and its small aligned residual remains consistent with a predominantly rigid phase error. Every saved η , ξ , and $G\xi$ value in these trajectories is finite; slope and spectral-band diagnostics remain bounded without an accelerating mid/high-band surge. The evidence therefore supports secular phase allocation, not a revived cascade or abrupt integrator failure.

This outcome does not mean that γ^2 is blind to differential error. The differential component is 3.8183903×10^{-6} , or 94.08% of C22’s mean train phase loss; only 2.4040945×10^{-7} is rigid. Rather, a corpus mean permits improvement of the aggregate differential statistic while redistributing rare tail risk. Consistently, the g_1 aligned p95 improves modestly from C21’s 0.14165 to C22’s 0.13773, while case 22 worsens from raw/aligned 0.39240/0.34147 to 0.41584/0.36167 and case 24 remains outside its gate.

The completed common-initial-state discriminator gives a sharper warning. Using the exact implemented Gaussian projection and energy floor, the C20 frame-zero differential scores are: The

Case	Differential rank	γ_{diff}^2	Final geometry
Tanaka g_0 case 27	1	$2.24346294649 \times 10^{-3}$	Predominantly rigid
Tanaka g_1 case 24	7	$8.62902609365 \times 10^{-6}$	Two-crest differential
Tanaka g_1 case 22	9	$2.58167939588 \times 10^{-6}$	Three-crest differential

Table 9: C20 frame-zero ranking under the implemented soft spatial-variance component.

statistic does detect cases 24 and 22 in the upper truth-state tail, but case 27 is about 260 times larger than case 24. Moreover, 97.47% of case 27’s implemented γ^2 is classified as differential even though a single rigid shift removes 97.78% of its C22 final squared elevation error. C22 preserves ranks 1/7/9 for cases 27/24/22, and all three frame-zero scores increase slightly despite the large rollout improvement of case 27. The soft spatial variance is therefore responding primarily to within-crest variation on the narrow single pulse; it is not a clean classifier of inter-crest phase.

Consequently, a naive fourth moment or unnormalized CVaR applied to the current γ_{diff}^2 should be rejected: it would further overweight the already-repaired single-crest case. Before another training run, formulate and validate either a coherent packet/crest-speed statistic or a capped, sample-balanced tail statistic over held-out truth trajectories. The new observable must rank the growing aligned errors of cases 22 and 24 without being dominated by case 27. If no truth-state statistic predicts that growth, then sparse signed one-substep tubular labels are justified as a response to off-trajectory covariate shift. This ordered test separates target design from data support before spending another optimization run.

C22 gives the best case-27 result and the best g_0 p95, exactly the direction predicted by the missing phase-sensitivity factor. It also slightly improves case 24 and the g_1 aligned p95 relative to C21. However, case 24 still fails its pre-registered raw/aligned gates of 0.50/0.15, and the g_1 aligned p95 of 0.13773 fails the approximately 0.107 gate. The combined raw and aligned medians regress relative to both controls; raw p95 also regresses. C22 therefore passes the stability, coefficient, H1/modal, case-27, and raw-threshold gates, but fails the multi-crest and distributional acceptance criteria.

The correct decision is *partial mechanism acceptance and checkpoint rejection*. C21 remains the promoted best aggregate accuracy/stability checkpoint, and C20 remains the clean loss-ablation control. Because C22 failed the decisive Tanaka gate, an eight-family non-Tanaka evaluation is not warranted: it cannot reverse the failed replacement decision and would spend substantial compute without resolving the mechanism. The result does not invalidate Eq. (35); rather, it shows that minimizing its corpus mean can improve the most phase-sensitive rigid tail while moving typical and differential errors. If this direction is continued, the next step is offline formulation and validation of either a coherent packet/crest-speed statistic or a capped, sample-balanced tail statistic over held-out truth trajectories. A naive fourth moment or unnormalized CVaR of the current spatial-variance component is specifically contraindicated by Table 9. Sparse signed one-substep tubular training remains the fallback only if no truth-state observable predicts the subsequent differential growth and the risk instead blooms on learned states.

A full Zakharov-vector projection onto the translation generator is useful as an offline diagnostic, but it is not the first training loss. The only learned quantity is q ; the ξ_t error is a deterministic nonlinear reuse of that same q , so a full-vector term supplies no independent label and requires a less canonical state-space metric. Equation (35) has the more direct first-order relation to the reported elevation error and adds neither a network evaluation nor inference-time work.

10.7 Stagewise modal-phase sufficiency oracle

Correlation of a truth-state loss with terminal error does not establish that the selected defect component is causally sufficient. We therefore performed a privileged stagewise oracle experiment on C21. At a current oracle state $z = (\eta, \xi)$, let

$$d(z) = q_6(z) - q_\theta(z), \quad q(\eta, \xi) = G(\eta)\xi,$$

where q_6 is the order-six, pad-eight reference evaluation and q_θ is the C21 evaluation at the *same* state. For every active mode $0 < |k| \leq 128$, define the real modal phase projection

$$\widehat{\Pi_\phi d_k} = i\widehat{\eta_k} \frac{\text{Im}(\widehat{d_k} \widehat{\eta_k})}{|\widehat{\eta_k}|^2}, \quad q_{\text{or}} = q_\theta + \Pi_\phi d. \quad (37)$$

Modes whose elevation energy is below 10^{-12} of the retained maximum are excluded. Equation (37) supplies exactly the same-state DNO-defect component tangent to the phase circle $\text{span}_{\mathbb{R}}\{i\widehat{\eta_k}\}$; it

leaves the radial modal component of the learned defect untouched. The correction is recomputed at every production GL2 stage using inner step 0.01, four Picard iterations, and the standard 0.25 filter.

This is a causal diagnostic, not a proposed inference method. In particular, the deployed analytic Craig–Sulem backbone remains exactly $G_0 + G_1$; the order-six evaluation appears only in the oracle reference and is neither added to the architecture nor used in ordinary rollout.

The complete fixed six-tail panel gives:

Case	C21 raw	C21 aligned	Oracle raw	Oracle aligned
4	0.276834	0.008247	0.149740	0.005570
27	0.336422	0.030498	1.024735	0.042920
2	0.374177	0.008365	0.243942	0.007618
11	0.345502	0.013085	0.113941	0.005149
22	0.392402	0.341471	0.244145	0.087457
24	0.659088	0.245457	0.625278	0.119105

Table 10: Final relative elevation errors for the fixed-six stagewise modal-phase oracle at $T = 200$. Case labels are local to their Tanaka family.

All six oracle trajectories remain finite. Four cross below the raw-error threshold 0.25, and the maximum aligned error falls from 0.341471 to 0.119105. The raw median improves from 0.359839 to 0.244043. Nevertheless, the raw counts above 0.25/0.50/0.75/1.00 change from 6/1/0/0 to 2/2/1/1: the intervention redistributes rather than eliminates tail risk.

The two counterexamples are complementary. For case 24, the aligned error falls by 51.5%, but the raw error falls by only 5.1%; the best-fit displacement changes only from 5.59 to 5.37 grid points. Its terminal modal phase component is 76.1% of the same-state DNO defect, so this failure cannot be dismissed as an inactive projection. Case 27 is more decisive: raw error increases from 0.336422 to 1.024735 and displacement from 1.69 to 6.22 grid points. At times 20, 40, 80, 120, 160, 200, its oracle raw errors are 0.0105, 0.0437, 0.2038, 0.4513, 0.7498, 1.0247, respectively. Thus the reversal is a smooth secular drift, not a late discontinuity. Throughout both cases, Hamiltonian drift remains bounded and the terminal $k = 64, \dots, 128$ spectrum stays close to truth.

The oracle therefore distinguishes causal contribution from causal sufficiency. The same-state modal phase component contributes materially to the aligned and aggregate error, but correcting it alone leaves the radial defect free to alter the state. The exact DNO response to that altered state can then regenerate or reverse the phase drift. For a multi-packet profile, a local tangent defect $a_1\eta_{1x} + a_2\eta_{2x}$ is also not generally parallel to $i\hat{\eta}_k$ when $a_1 \neq a_2$; genuine differential packet translation is therefore split between the global modal phase and radial subspaces. A modal phase-rate loss remains mathematically meaningful, but the complete oracle rejects it as a sufficient replacement for a localized packet-translation or coupled finite-horizon sensitivity.

Cross-case transfer falsifies a universal reduced projection. The same five-arm experiment on the predominantly rigid Tanaka- g_0 case 27 reverses the case-24 ordering:

The complement of the modal phase projection is the best reduced arm for case 24, whereas the width- $4h$ localized projection is the best reduced arm for case 27. Neither transfers: the case-24 winner leaves case 27 at raw error 0.5355, while the case-27 winner leaves case 24 at 0.4744. Moreover, modal-phase and width- h correction make case 27 worse than one, despite supplying an exact component of the reference defect. The width- $4h$ case-27 trajectory is not uniformly repaired either: its raw error reaches 0.2819 at $t = 120$ before decreasing to 0.1346.

Correction	Case 24		Case 27	
	Raw	Aligned	Raw	Aligned
None, C21	0.659088	0.245457	0.336422	0.030498
Modal phase $\Pi_\phi d$	0.625276	0.119104	1.024433	0.042911
Localized translation, width h	0.508067	0.225268	1.079470	0.043482
Localized translation, width $4h$	0.474397	0.151743	0.134637	0.024499
Complement $d - \Pi_\phi d$	0.137739	0.136753	0.535542	0.027532
Full defect d	2.254×10^{-8}	2.254×10^{-8}	1.937×10^{-8}	1.844×10^{-8}

Table 11: Terminal relative elevation errors for two C21 Tanaka tails under stagewise components of the same-state order-six defect $d = q_6 - q_\theta$.

This nonmonotonicity is expected once the defect components are propagated by the coupled nonlinear Zakharov dynamics. Orthogonality of instantaneous Fourier components does not imply orthogonality of their finite-time effects; the learned phase and amplitude defects can partially cancel after feedback through both η and ξ . Removing only one exact component may therefore expose the secular effect of the component left behind.

By contrast, full-defect correction shadows the reference to approximately 2×10^{-8} in both cases. This validates the GL2 implementation for these trajectories and identifies the common missing ingredient as uniformly better approximation of the full complex DNO response, not a universal scalar phase coordinate. Since the existing operator objective already supervises that full defect, the clean next test is a modest increase in learned-residual capacity followed by a full structural retraining with the established value, mode-balanced, and Hadamard objectives. This changes neither the analytic ceiling nor the inference principle: the deployed analytic backbone remains exactly $G_0 + G_1$, with all higher-order behavior represented by the learned $O(\eta^2)$ residual. The oracle motivates this experiment but does not itself establish that capacity, rather than optimization, is the limiting factor.

11 Alternative explanations, objections, and falsifiers

This section states the questions most likely to arise in a rigorous oral discussion and the limits of the current answers.

11.1 Could the data still contain subtle corruption?

Yes, in the weak logical sense that no finite audit proves every stored number correct. What has been ruled out is the form of corruption needed to explain the observed failure mechanism. Random and source-stratified scans found no nonfinite base rows; sign and spectral heuristics did not reveal a population aligned with the failing cases; and, most importantly, the learned DNO has small value error when evaluated on the exact failing truth states. A corrupted label population severe enough to seed the reported sideband directly should normally appear as increased error on those states. Instead, the large error appears when the input is moved in the rollout’s transverse direction.

A useful remaining audit would compare independently regenerated order-six labels against stored v9 labels on a stratified sample, including source, depth, steepness, and high-band-energy tails. Such an audit would strengthen the evidence for the stored labels but would not replace the already measured shape identity.

11.2 Does small value error plus large derivative error necessarily cause a NaN?

No. A large local derivative defect is a mechanism and risk factor, not by itself a theorem of finite-time blow-up. The causal case is empirical: the defect is present at the failing states, the realized secant direction grows, its spectrum matches the subsequent cascade, and analytic restoration of the first-order null form removes all NaNs in the 32-case test. A formal theorem would require stability bounds for the coupled semi-discrete system and a uniform neighborhood estimate for $D\mathcal{G}_\theta$, neither of which is currently available.

11.3 Could GL2 still be the primary issue?

The fixed number of Picard iterations is certainly where nonfiniteness is finally emitted, and a more robust nonlinear solver could extend some trajectories. It is unlikely to be the primary issue in the audited cases because the model-induced mid-band response is already orders of magnitude above its truth-state bias before the stage residual fails; substep refinement and padded-RHS replay delay failure only slightly; and healthy cases use the same integrator without trouble. A definitive follow-up could solve selected near-terminal stages with Newton–Krylov to high tolerance. If the converged implicit map returned to a healthy spectral state, the present conclusion would need revision. If it advanced the already contaminated state, it would confirm the current ordering.

11.4 Why not learn the whole Hamiltonian and differentiate it?

That is a principled alternative. A scalar Hamiltonian network with $\eta_t = \delta_\xi H_\theta$ and $\xi_t = -\delta_\eta H_\theta$ would be Hamiltonian by construction. The tradeoff is computational: rollout requires derivatives of a nonlocal learned energy, training DNO values can involve mixed derivatives, and encoding the known linear dispersion and Craig–Sulem structure is less direct. C16 keeps the efficient DNO interface while enforcing the identity that makes that interface Hamiltonian. If C16 fails despite a substantially reduced measured shape defect, a direct energy architecture becomes the most principled next architectural family.

11.5 Why not impose a global Lipschitz or spectral norm bound?

A uniform bound could control amplification but would not distinguish physical from unphysical high-frequency response. The DNO is an order-one nonlocal operator; its norm naturally scales with wavenumber, and nonlinear wave dynamics legitimately transfer energy across modes. A naive global Lipschitz constant or hard cutoff risks destroying exactly the dynamics we are trying to approximate. The Hadamard identity is selective: it constrains how the large terms must cancel, rather than requiring every term to be small.

11.6 What result would falsify the C16 mechanism?

The strongest falsifier would be: the final C16 checkpoint reduces the held shape defect by two orders of magnitude and preserves the same-sign G_1 null form, yet the same cases still develop the same pre-cascade secant gain and spectral signature. That would show that the diagnosed identity is not the controlling mechanism. Other negative outcomes are less decisive:

- If the shape defect does not improve, the loss estimator, weight, or 16-epoch budget failed to enforce the hypothesis; the hypothesis itself was not tested strongly.
- If the defect improves but one-step accuracy is poor, the structural constraint may be correct but the residual architecture or optimization is inadequate.

- If NaNs disappear but p95 error becomes unacceptable, C16 is a stability result but not a useful surrogate.

A Chronological experiment map

This appendix is deliberately redundant with `EXPERIMENTS.md`. Its purpose is to make the logic reconstructible without memorizing labels.

Handle	Intervention	Outcome	What it taught us
v3–v10	Full-feature CS-DNO baselines; progressively broader clean data and a double-precision rollout harness.	Much better one-step and median rollout accuracy than generic FNOs, but persistent rare Tanaka NaNs (v10: 4/32).	The Craig–Sulem inductive bias is valuable, but pointwise accuracy and broad data coverage do not ensure rollout stability.
C1	Light GL2 stage-tangent matching fine-tune.	Tanaka 4/32 to 3/32 NaNs; Benjamin–Feir 2/31 to 0/31.	Off-manifold sensitivity is a real lever, though the indirect proxy is incomplete.
C2, first launch	Stage match plus gain, but resumed a tied checkpoint into an untied architecture.	32/32 NaNs in both decisive regimes.	Invalid experiment; motivated strict resume architecture and parameter-count guards.
C2, corrected	Same intended loss with tied architecture preserved; final checkpoint evaluated after the auxiliary loss acted.	Tanaka 3/32 NaNs, median 0.00907; BF 0/31, median 0.00735.	Final versus best checkpoint matters for auxiliary-loss fine-tunes; still short of zero NaNs.
GL2 guard	Picard residual detection, attempted halving, then per-sample masking.	Detected but did not reduce NaN counts; early fleet-wide logic could couple one bad sample to the whole batch.	GL2 noncontractivity is downstream; batch-local safety logic must remain per sample.
Block-7 ablation	Zero an apparently amplifying learned block.	Tanaka NaNs rose from 4/32 to 28/32; median error rose 62-fold.	Trained blocks form a fragile compensating sum; post hoc deletion is not an architectural study.
v11-D	80-fold smaller model plus cutoff, stage regularizer, and PSD hinge.	Same Tanaka NaN count as baseline and 12–20-fold median regressions.	A kitchen-sink run is hard to interpret and reduced capacity was not a cure.
v11-E	Full capacity with hard learned-block k -cut isolated.	Did not establish a useful zero-NaN/accuracy tradeoff.	Brute-force band removal is not the preferred mechanism.
C6/C7	Model-harvested pre-cascade hard negatives.	Harmful or non-general fine-tunes.	Contaminated off-manifold packs can feed the model’s own failure back into training.

Handle	Intervention	Outcome	What it taught us
C8/C10	Increasingly matched stage-gain regularization, with C10 using the production filter and $k = 64, \dots, 128$ target band.	C10 reached the best pre-C16 count, 2/32 Tanaka and 0/31 BF NaNs, but regressed some medians/tails.	The diagnosed spectral band matters; indirect regularization still does not restore the defining operator identity.
C9	Stronger soft full-output high-band envelope penalty.	Penalty diagnostic barely moved; Tanaka remained 3/32 NaNs and accuracy regressed.	Simply increasing this penalty weight is not an effective axis.
C10 + runtime limiter	Cap learned $G\xi$ high-band output during rollout.	Tanaka stayed 2/32 NaNs; BF p95 worsened.	The operational guard does not reach zero and can damage tails.
C11/C13	Clean truth-generated hard-negative packs near selected remaining failures.	Training was feasible; C13 evaluation was superseded before a final claim.	Useful as a local learnability probe, not a universal or satisfying paper-level mechanism.
C12	One-step pushforward/filtering fallback prepared from C2.	Not promoted after the stronger structural diagnosis.	Rollout-state training remains an option, but it mixes integrator and operator effects.
C14	One-block, checkpoint-carrying full-output cap.	Best finite checkpoint: 3/32 Tanaka NaNs, 6/32 divergences, median 0.162, p95 1.253; training later became nonfinite.	The cap has a mid-band/floor loophole and severe accuracy cost; one-block capacity was also a confounder.
C15	Eight-block capacity control for the C14 cap.	Interrupted before epoch 1 and before any checkpoint.	Inconclusive as a capacity ablation, but not worth relaunching because the cap mechanism itself was analytically rejected.
Root-cause audit	Frame alignment, realized secants, automatic-differentiation shape check, transfer kernel, analytic $G_0 + G_1$, and padded-RHS replay.	Small truth-state value error coexists with $O(1)$ shape defect; exact $G_0 + G_1$ is stable; RHS padding delays rather than cures failure.	The DNO shape derivative/Hamiltonian compatibility is the primary missing constraint.
C16	Fresh exact $G_0 + G_1$, full self-adjoint $O(\eta^2)$ residual, randomized Hadamard loss, and padded RHS.	Clean 40-epoch replay and fixed ten-family evaluation give 0/320 nonfinite trajectories; only two finite Tanaka phase divergences remain.	Direct structural treatment removes the NaN mechanism, but stability does not by itself give adequate phase accuracy.

Handle	Intervention	Outcome	What it taught us
C20	Fresh C16 architecture with mode-balanced supervision, corrected local translation-tangent loss at weight 10, and Hadamard loss retained.	0/320 nonfinite predictions; full-suite median/p95/mean 0.007201/0.16082/0.03464, but one rigid Tanaka error remains above one.	Weak-mode accuracy improves substantially; the remaining extreme error is a neutral translation, not a spectral cascade.
C21	One low-learning-rate epoch from C20 with only the old localized-tangent coefficient increased from 10 to 100.	0/320 nonfinite and zero final divergences; full-suite median/p95/mean 0.006866/0.15369/0.03205. Case 27 improves strongly, while two-crest case 24 worsens.	Best aggregate checkpoint, but a larger mean phase coefficient redistributes rigid and differential speed error rather than eliminating both.
Phase audit	Continuous alignment, local crest alignment, spectral and Hamiltonian checks, and conversion from local speed error to relative state error.	Derives $\gamma^2 = gh\kappa^2\ell_{\tan}$; its frame-zero Pearson log-score/log-final-error correlation is 0.737 versus 0.650 for the old loss, while raw-scale correlation does not improve.	Motivates an isolated phase-sensitivity normalization test before any new data, architecture, or full retraining.
C22	One full-v9 epoch from C20 with the old tangent loss off and the derived mean γ^2 loss at calibrated weight 10; phase validation included.	0/64 Tanaka nonfinite predictions; case 27 improves to 0.18047/0.02688 and the raw threshold counts match C21, but case 24 remains 0.63952/0.23662 and aggregate medians regress.	Partial confirmation of the sensitivity mechanism, but reject as a replacement for C21; a mean phase loss does not control the rare differential tail.

B Implementation and correctness checks

The implementation was not accepted on training loss alone. The following independent checks passed before the full launch:

- five architecture tests: legacy order-one identity, zero order-two value and first variation, quadratic near-zero scaling, self-adjointness, and isolation of double precision to analytic G_1 ;
- seven Hadamard-loss tests: projected Sobolev energy, probe scaling and bandwidth, exact flat-surface identity, detection of missing G_1 , production normalization/mean behavior, JIT and mixed-precision gradients, and invalid-configuration handling;
- two dedicated padded-RHS tests: equality on bandlimited products and removal of high-frequency quadratic aliasing;
- fourteen existing stage-regularizer tests after routing all nonlinear right-hand sides through the same padded helper; and

- a 59-step two-GPU smoke with finite checkpoints and validation.

Reproducibility record

The detailed chronological record is `EXPERIMENTS.md`. Frame-level statistics and paths to the aligned NPZ archives are recorded in `notes/nan_root_cause_investigation_20260709.md`. The C16 launcher is `scripts/launch_c16_modal_h100x2.sh`; model, Hadamard-loss, and dealiased-RHS invariant tests are stored alongside their implementations. The C21 checkpoint and Tanaka evaluation are under `outputs/c21_tangent_w100_from_c20_20260714_171442`; its eight-family acceptance evaluation is the subdirectory beginning `eval_final_guarded_non_tanaka_suite`. The frame-by-frame phase audit and complete derivation of Eq. (35) are in `notes/principled_translation_error_solution_20260714.md`. The C22 run is `outputs/c22_phase_growth_w10_from_c20_20260715_012811`; its matched Tanaka evaluation is the subdirectory `eval_final_soliton_spectral_guard_20260715_023013`.

References

- [1] L. Xu and P. Guyenne, “Numerical simulation of three-dimensional nonlinear water waves,” *Journal of Computational Physics*, vol. 228, pp. 8446–8466, 2009, doi:10.1016/j.jcp.2009.08.015.